

NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

32-bit Hi-Fi DAC

The BD34352EKV digital-to-analogue converter (DAC) IC supports playback of high-resolution sound sources



in high-fidelity audio equipment. Designed to maximally extract and convert high-resolution digital audio data to analogue audio signals, audio DAC chips are one of the most important components for determining the quality in audio equipment. The BD34352EKV maintains a natural flat sound while at the same time an advanced digital filter is included. This leads to superior numerical performance (S/N ratio of 126dB, THD + N of -112dB) of the DAC chip, making it ideal for high-end audio equipment.

ROHM Semiconductor
www.rohm.com

Smallest SD card level translator

The NXS0506UP is the world's smallest secure digital (SD) card level translator IC. Housed in a 16-bump wafer-level

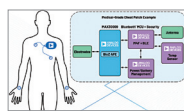


chip-scale package, the SD 3.0-compliant bi-directional dual voltage level translator features integrated auto-direction control, EMI filter, and IEC 61000-4-2 ESD protection. Designed to operate at clock frequencies of up to 208MHz and data rates of up to 104Mbps, it also features integrated pull-up and pull-down resistors to reduce the BOM cost. Devices provide direct connection to the latest host CPU/SOC ASICs and are easy to use because DATA and CMD channels are swappable without impacting functionality.

Nexperia
www.nexperia.com

Low-power BioZ analogue front-end

The MAX30009 is a low-power, high-performance bioimpedance (BioZ) analogue front-end (AFE) that reduces the size and extends the life of BioZ



remote-patient monitors. For developers of small, battery-powered, continuously wearable devices, this AFE on a chip offers clinical-grade vital sign measurements of bioimpedance analysis for patient health assessment for wellness wearables and medical-grade patches. The MAX30009 has a range of options to enable use-case power optimisation, thus extending the operational life of BioZ wearables. It reduces power consumption by 62% and has 30% smaller size than other alternatives, allowing designers to help users measure their vital signs more comfortably.

Analog Devices
www.analog.devices

High-precision pressure transmitters

The AVD series of pressure transmitters are available in three versions



for different pressure measurement ranges, including 0 to 16mbar, 0 to 100mbar, and 0 to 7bar. The series offers high precision levels for low-pressure ranges, such as $\pm 1.75\%$ full scale with a pressure range of 100mbar. The compact piezoresistive pressure transmitters based on MEMS technology are suitable for the pressure measurement of air and non-aggressive gases. In addition to dif-

ferential pressure measurement, they can also measure over-pressure in a temperature range of -20°C to $+70^{\circ}\text{C}$.

TDK Corporation
www.tdk.com

Crankshaft and camshaft sensor

The ATS16951 and ATS16351 are two new state-of-the-art giant magnetoresistance (GMR) crankshaft and camshaft sensors, respectively.



They provide manufacturers with a single-vendor solution that's ideal for hybrid vehicle engines, with uses extending to two-wheelers, off-road vehicles, and application designs requiring extended air gap performance. Both sensors help engine designers reduce system complexity, cost, and energy consumption, thus boosting efficiency and minimising carbon footprints. The ATS16951 combines electrical requirements and package size of legacy solutions with sensing location flexibility. The ATS16351 is highly programmable and allows manufacturers to optimise the TPOS performance across a variety of engine designs.

Allegro MicroSystems
www.allegromicro.com

Automotive Gen 4 PCIe switches

The Gen 4 automotive-qualified PCIe switches, which include Switchtec PFX, PSX, and PAX switch solutions,



provide cutting-edge compute interconnect capabilities for advanced driver assistance systems (ADAS). Switchtec Gen 4 PCIe switches offer a high-speed interconnect that supports distributed, real-time, safety-critical data processing in ADAS architectures. PCIe is emerging as the preferred compute interconnect solution for the automotive industry.